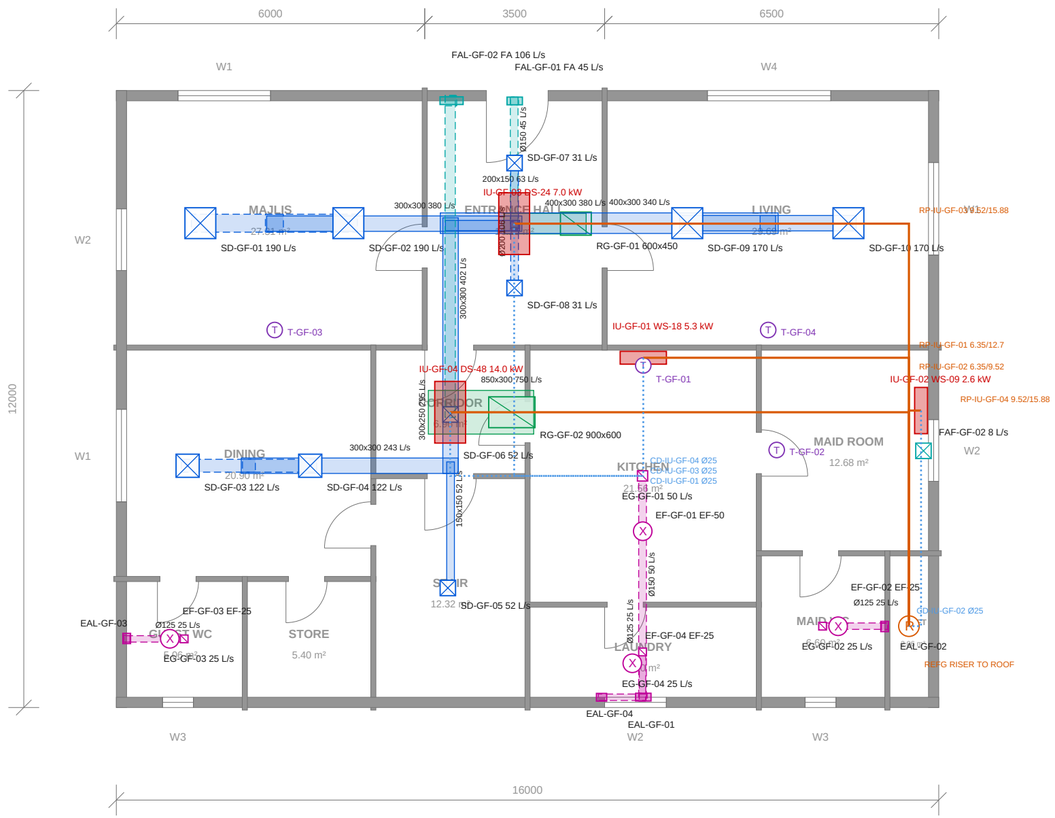
GROUND FLOOR PLAN





WINDOW SCHEDULE

W1	1800 x 1500	ALUMINIUM, DOUBLE GLAZED, SILL 900
W2	1200 x 1500	ALUMINIUM, DOUBLE GLAZED, SILL 900
W3	600 x 600	ALUMINIUM, DOUBLE GLAZED, SILL 900
W4	2400 x 2100	ALUMINIUM, DOUBLE GLAZED, SILL 900

PROJECT: SAMPLE VILLA - RIYADH

DRAWING: GROUND FLOOR PLAN

SCALE 1:100 @ A3

DWG NO. A-101 REV 0

ARCHITECT: SAMPLE ARCHITECTS (TEST DATA)

PRELIMINARY DESIGN - FOR COORDINATION ONLY. To be checked and stamped by a qualified HVAC engineer before construction.

HVAC LAYOUT - GROUND FLOOR



North 20° from page-up

Scale 1:100 (as architectural sheet)

LEGEND

- Supply duct (size, flow) / flex dashed
- Supply diffuser SD
- Return duct
- Return grille RG
- Fresh air duct / louver FAL
- Exhaust duct / grille EG / fan EF
- Refrigerant pipe (liquid/gas mm)
- Condensate drain
- Chilled water pipe
- Indoor / outdoor unit
- Thermostat / controller

EQUIPMENT ON THIS SHEET

IU-GF-01	WS-18	5.3 kW	250 L/s Z-GF-01
IU-GF-02	WS-09	2.6 kW	140 L/s Z-GF-02
IU-GF-03	DS-24	7.0 kW	380 L/s Z-GF-03
IU-GF-04	DS-48	14.0 kW	750 L/s Z-GF-04
EF-GF-01	EF-50	50 L/s	GF-06
EF-GF-02	EF-25	25 L/s	GF-09
EF-GF-03	EF-25	25 L/s	GF-11
EF-GF-04	EF-25	25 L/s	GF-13

NOTES

- Duct sizes W x H mm (clear internal); Ø = round.
- Ducts at high level in ceiling void; coordinate with beams.
- Door undercut 20 mm / transfer grilles for return air.
- Refrigerant/condensate routes are indicative.

PROJECT: Sample Villa - Riyadh

LOCATION: Riyadh, Saudi Arabia

SHEET: HVAC LAYOUT - Ground Floor

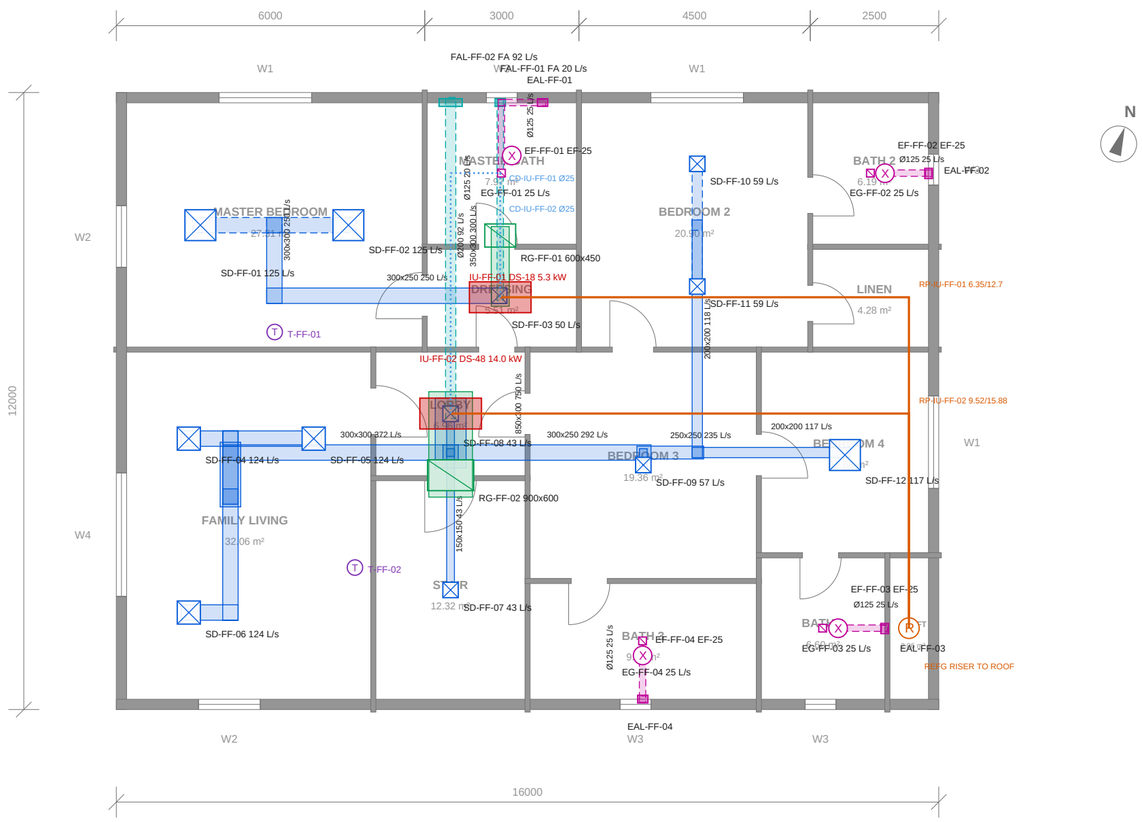
SHEET NO. M-101 REV P0 2026-10-15

SCALE: 1:100 CODES: Saudi Building Code + ASHRAE

PREPARED WITH hvac-designer skill (automated)

PRELIMINARY DESIGN - FOR COORDINATION ONLY. To be checked and stamped by a qualified HVAC engineer before construction.

FIRST FLOOR PLAN



WINDOW SCHEDULE

W1 1800 x 1500 ALUMINIUM, DOUBLE GLAZED, SILL 900
W2 1200 x 1500 ALUMINIUM, DOUBLE GLAZED, SILL 900
W3 600 x 600 ALUMINIUM, DOUBLE GLAZED, SILL 900
W4 2400 x 2100 ALUMINIUM, DOUBLE GLAZED, SILL 900

PROJECT: SAMPLE VILLA - RIYADH

DRAWING: FIRST FLOOR PLAN

SCALE 1:100 @ A3

DWG NO. A-102 REV 0

ARCHITECT: SAMPLE ARCHITECTS (TEST DATA)

PRELIMINARY DESIGN - FOR COORDINATION ONLY. To be checked and stamped by a qualified HVAC engineer before construction.

HVAC LAYOUT - FIRST FLOOR

N

North 20° from page-up
Scale 1:100 (as architectural sheet)

LEGEND

- Supply duct (size, flow) / flex dashed
- Supply diffuser SD
- Return duct
- Return grille RG
- Fresh air duct / louver FAL
- Exhaust duct / grille EG / fan EF
- Refrigerant pipe (liquid/gas mm)
- Condensate drain
- Chilled water pipe
- Indoor / outdoor unit
- Thermostat / controller

EQUIPMENT ON THIS SHEET

IU-FF-01 DS-18 5.3 kW 300 L/s Z-FF-01
IU-FF-02 DS-48 14.0 kW 750 L/s Z-FF-02
EF-FF-01 EF-25 25 L/s FF-01
EF-FF-02 EF-25 25 L/s FF-02
EF-FF-03 EF-25 25 L/s FF-12
EF-FF-04 EF-25 25 L/s FF-14

NOTES

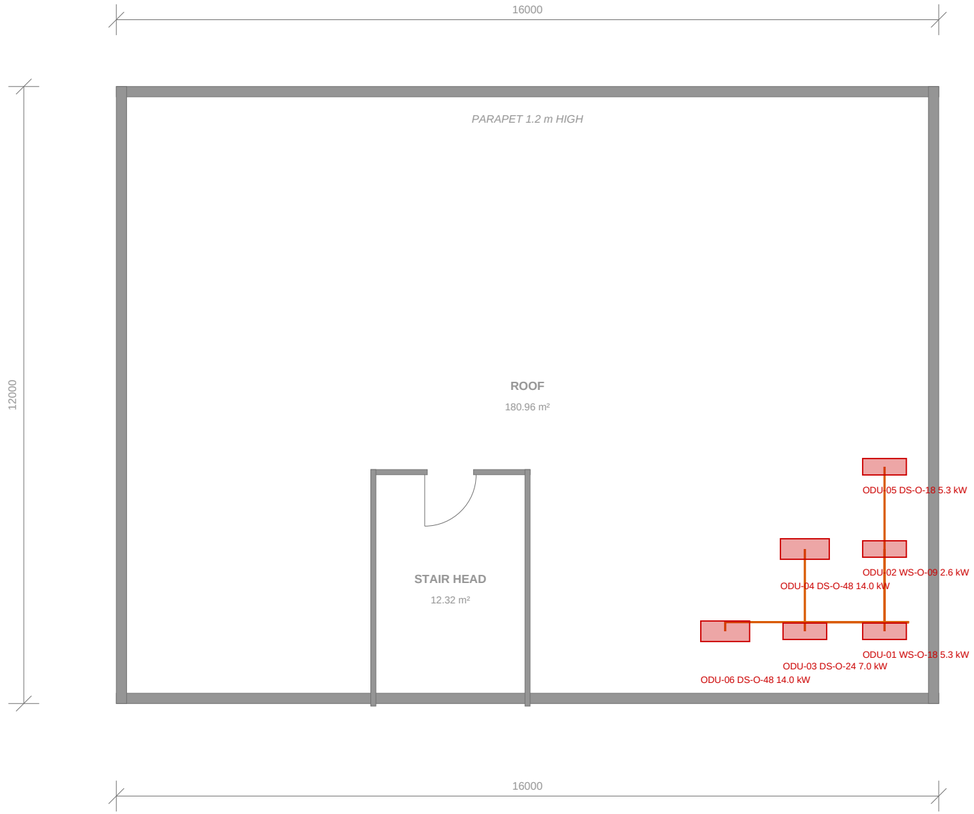
- Duct sizes W x H mm (clear internal); Ø = round.
- Ducts at high level in ceiling void; coordinate with beams.
- Door undercut 20 mm / transfer grilles for return air.
- Refrigerant/condensate routes are indicative.

PROJECT: Sample Villa - Riyadh
LOCATION: Riyadh, Saudi Arabia
SHEET: HVAC LAYOUT - First Floor
SHEET NO. M-102 REV P0 2026-10-15
SCALE: 1:100 CODES: Saudi Building Code + ASHRAE
PREPARED WITH hvac-designer skill (automated)

PRELIMINARY DESIGN - FOR COORDINATION ONLY. To be checked and stamped by a qualified HVAC engineer before construction.

PRELIMINARY DESIGN - FOR COORDINATION ONLY. To be checked and stamped by a qualified HVAC engineer before construction.

ROOF PLAN



N

PROJECT: SAMPLE VILLA - RIYADH

DRAWING: ROOF PLAN

SCALE 1:100 @ A3

DWG NO. A-103 REV 0

ARCHITECT: SAMPLE ARCHITECTS (TEST DATA)

PRELIMINARY DESIGN - FOR COORDINATION ONLY. To be checked and stamped by a qualified HVAC engineer before construction.

HVAC LAYOUT - ROOF

N

North 20° from page-up

Scale 1:100 (as architectural sheet)

LEGEND

- Supply duct (size, flow) / flex dashed
- Supply diffuser SD
- Return duct
- Return grille RG
- Fresh air duct / louver FAL
- Exhaust duct / grille EG / fan EF
- Refrigerant pipe (liquid/gas mm)
- Condensate drain
- Chilled water pipe
- Indoor / outdoor unit
- Thermostat / controller

EQUIPMENT ON THIS SHEET

ODU-01 WS-O-18 5.3 kW serves IU-GF-01
ODU-02 WS-O-09 2.6 kW serves IU-GF-02
ODU-03 DS-O-24 7.0 kW serves IU-GF-03
ODU-04 DS-O-48 14.0 kW serves IU-GF-04
ODU-05 DS-O-18 5.3 kW serves IU-FF-01
ODU-06 DS-O-48 14.0 kW serves IU-FF-02

NOTES

- Duct sizes W x H mm (clear internal); Ø = round.
- Ducts at high level in ceiling void; coordinate with beams.
- Door undercut 20 mm / transfer grilles for return air.
- Refrigerant/condensate routes are indicative.

PROJECT: Sample Villa - Riyadh

LOCATION: Riyadh, Saudi Arabia

SHEET: HVAC LAYOUT - Roof

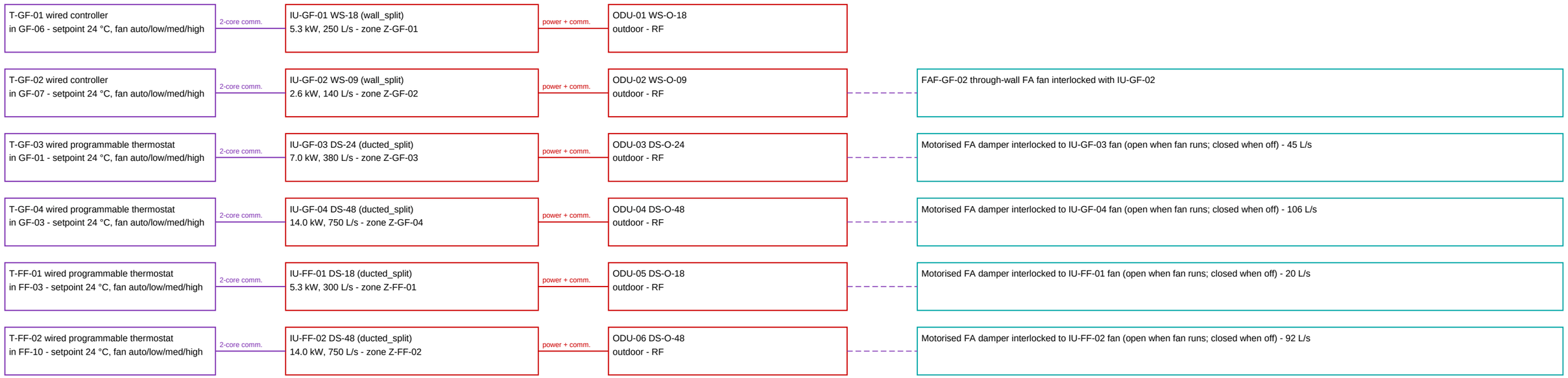
SHEET NO. M-103 REV P0 2026-10-15

SCALE: 1:100 CODES: Saudi Building Code + ASHRAE

PREPARED WITH hvac-designer skill (automated)

PRELIMINARY DESIGN - FOR COORDINATION ONLY. To be checked and stamped by a qualified HVAC engineer before construction.

HVAC CONTROLS SCHEMATIC AND EQUIPMENT SCHEDULE



EXHAUST FANS: switched with room lighting with 10 min run-on timer (bath/WC); kitchen fan on local switch; backdraft damper at each discharge.

Optional: central on/off + fault alarm to home automation / BMS via unit gateway (by specialist).

EQUIPMENT SCHEDULE

Tag	Zone	Type	Model	Cooling	Required	Airflow	ESP Pa	Heating	Power kW	Host / location
IU-GF-01	Z-GF-01	wall_split	WS-18	4.7 kW	3.6 kW	250 L/s	0	5.6 kW	1.61	GF-06
IU-GF-02	Z-GF-02	wall_split	WS-09	2.3 kW	0.9 kW	140 L/s	0	2.9 kW	0.76	GF-07
IU-GF-03	Z-GF-03	ducted_split	DS-24	6.2 kW	4.3 kW	380 L/s	75	7.6 kW	2.19	GF-02
IU-GF-04	Z-GF-04	ducted_split	DS-48	12.3 kW	9.1 kW	750 L/s	100	15.0 kW	4.52	GF-04
IU-FF-01	Z-FF-01	ducted_split	DS-18	4.7 kW	2.7 kW	300 L/s	75	5.8 kW	1.66	FF-04
IU-FF-02	Z-FF-02	ducted_split	DS-48	12.3 kW	9.7 kW	750 L/s	100	15.0 kW	4.52	FF-07
ODU-01	-	wall_split	WS-O-18	5.3 kW	-	-	-	-	1.61	RF
ODU-02	-	wall_split	WS-O-09	2.6 kW	-	-	-	-	0.76	RF
ODU-03	-	ducted_split	DS-O-24	7.0 kW	-	-	-	-	2.19	RF
ODU-04	-	ducted_split	DS-O-48	14.0 kW	-	-	-	-	4.52	RF
ODU-05	-	ducted_split	DS-O-18	5.3 kW	-	-	-	-	1.66	RF
ODU-06	-	ducted_split	DS-O-48	14.0 kW	-	-	-	-	4.52	RF
EF-GF-01	GF-06	exhaust fan ceiling	EF-50	-	-	50 L/s	80	-	0.025	GF
EF-GF-02	GF-09	exhaust fan ceiling	EF-25	-	-	25 L/s	60	-	0.015	GF
EF-GF-03	GF-11	exhaust fan ceiling	EF-25	-	-	25 L/s	60	-	0.015	GF
EF-GF-04	GF-13	exhaust fan ceiling	EF-25	-	-	25 L/s	60	-	0.015	GF
EF-FF-01	FF-01	exhaust fan ceiling	EF-25	-	-	25 L/s	60	-	0.015	FF
EF-FF-02	FF-02	exhaust fan ceiling	EF-25	-	-	25 L/s	60	-	0.015	FF
EF-FF-03	FF-12	exhaust fan ceiling	EF-25	-	-	25 L/s	60	-	0.015	FF
EF-FF-04	FF-14	exhaust fan ceiling	EF-25	-	-	25 L/s	60	-	0.015	FF

PROJECT: Sample Villa - Riyadh

LOCATION: Riyadh, Saudi Arabia

SHEET: CONTROLS SCHEMATIC & SCHEDULE

SHEET NO. M-901 REV P0 2026-10-15

SCALE: NTS CODES: Saudi Building Code + ASHRAE

PREPARED WITH hvac-designer skill (automated)

PRELIMINARY DESIGN - FOR COORDINATION ONLY. To be checked and stamped by a qualified HVAC engineer before construction.